

AQuaTomo: A Hardware-Agnostic Quantum State Tomography Suite

Niels Bohr
Quantum
Summer School

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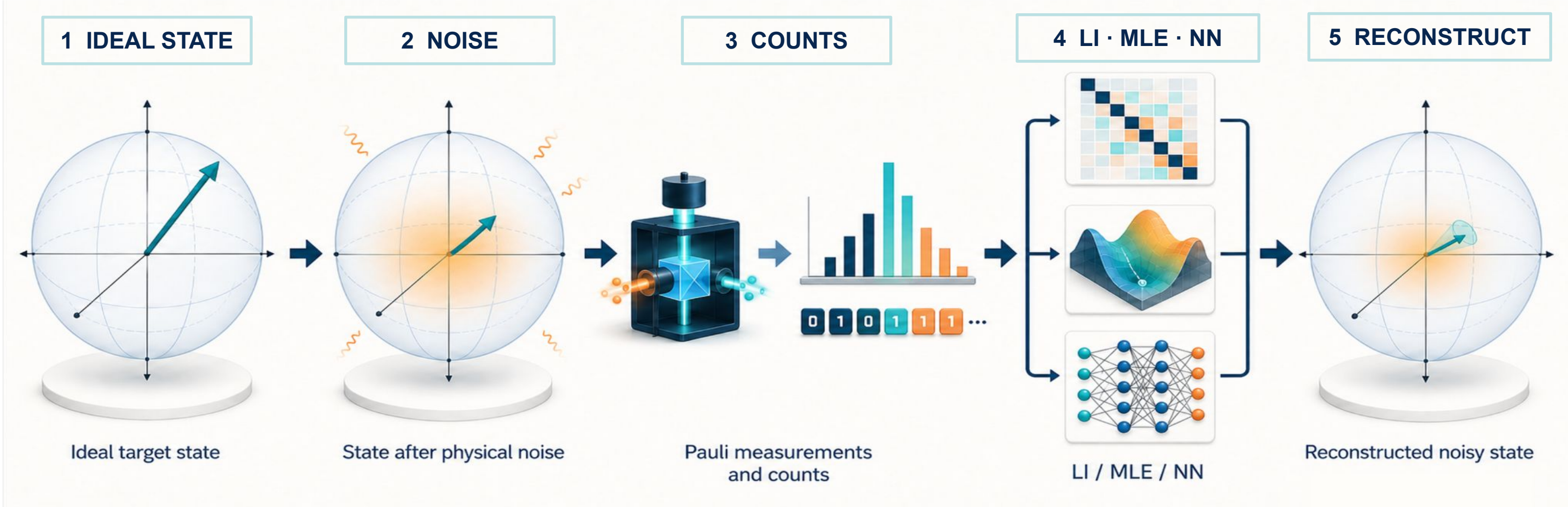
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PROBLEM & MOTIVATION

- Quantum State Tomography reconstructs a system’s density matrix from repeated measurements.
- Exponentially expensive:** Finite sampling and hardware noise can produce unphysical estimates, while $2^n \times 2^n$ density matrix makes QST exponentially expensive.
- Scalable tomography therefore requires physical reconstruction methods and hardware-agnostic CPU/GPU acceleration.

METHODOLOGY

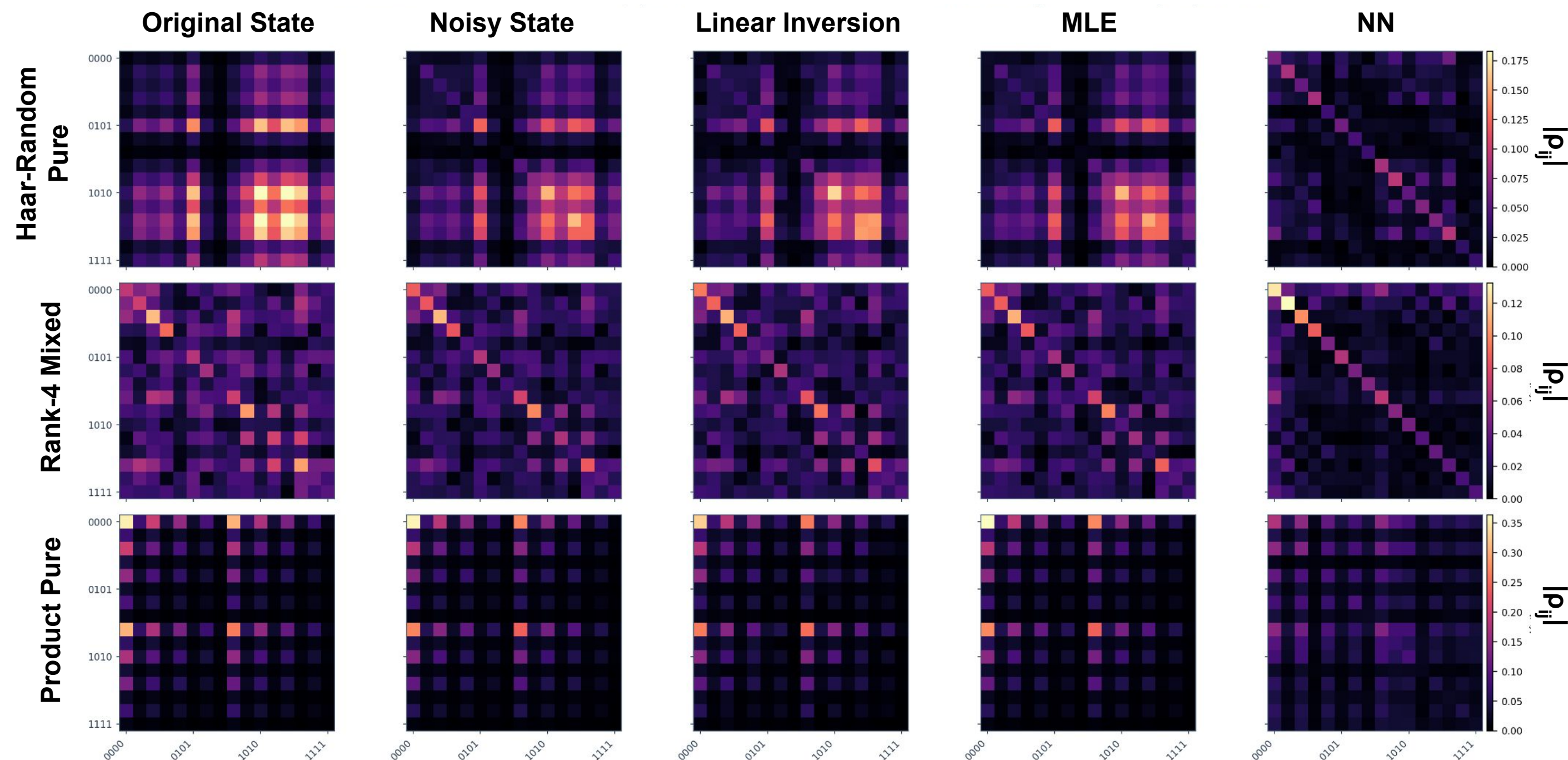


Architecture & State Simulation

- Hardware-agnostic:** Array-API routing runs the same mutation-free numerical code natively with NumPy, CuPy, or JAX on CPUs and GPUs.
- Realistic tomography:** Counts are sampled from all (3^n) local Pauli settings using finite-shot multinomial statistics, ensuring informational completeness.

Four-qubit density-matrix reconstruction

Local depolarizing $p=0.04$ followed by amplitude damping $\gamma=0.06$; 1,000 shots per each of 81 Pauli settings. LI, MLE, and NN target the noisy state; mitigation is not applied.



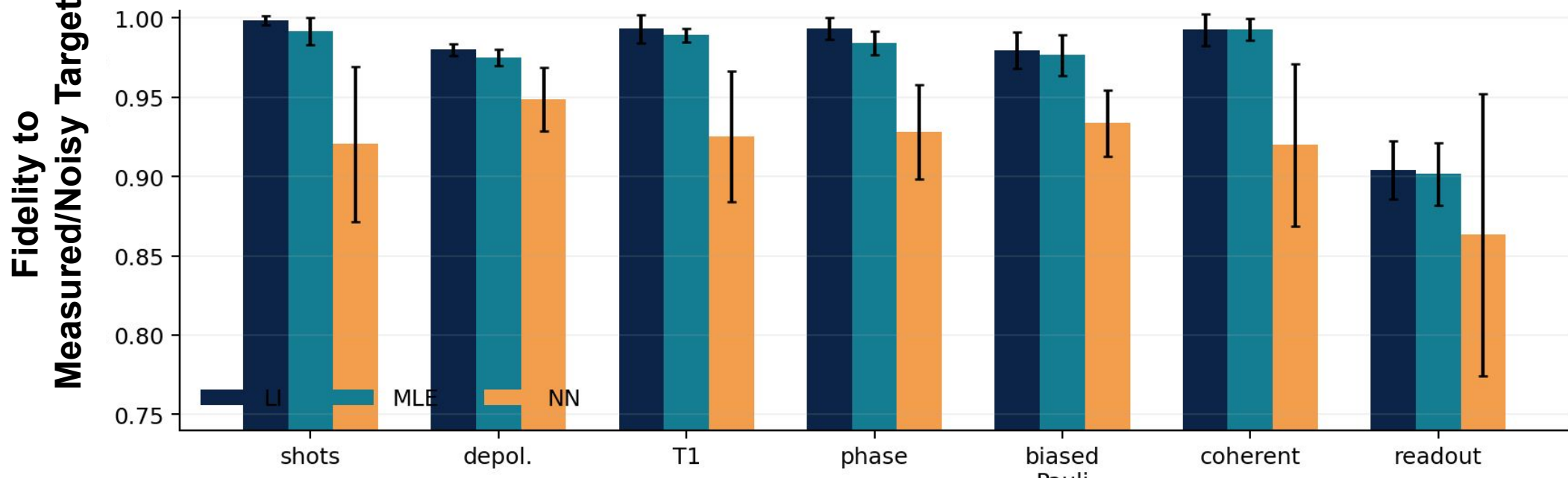
Linear Inversion (LI): Fast transparent baseline; usually outside state space

Maximum Likelihood Estimation (MLE): Best mean HS result in this validation run

Neural Network + Cholesky Reconstruction: Physical and fast to evaluate, but undertrained relative to the literature protocol

State	Reconstruction Method	Mean HS Distance	Mean Fidelity	Physical Fraction
Haar Pure D_{HS}	LI	0.127	-	-
	MLE	0.049	0.940	100%
	NN	0.665	0.527	100%
Rank-4 Mixed	LI	0.130	-	-
	MLE	0.070	0.927	100%
	NN	0.353	0.706	100%
Product Pure	LI	0.120	-	-
	MLE	0.034	0.981	100%
	NN	0.495	0.767	100%

Two-qubit Reconstruction Under Unmodeled Noise (500 shots/setting)



CONCLUSIONS

- A hardware-agnostic framework for state generation, measurement simulation, and state reconstruction was implemented and validated with both NumPy and JAX.
- Dense MLE scales exponentially: M1 CPU runtime increased from (0.00255) s at 1 qubit to (0.48050) s at 5 qubits.
- Future work will benchmark JAX/CuPy on HPC cluster GPUs and use low-rank or tensor-network representations beyond dense ($O(4^n)$) tomography.

REFERENCES

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[2] Consortium for Python Data API Standards, “Python Array API Standard: Purpose and Scope,” https://data-apis.org/array-api/latest/purpose_and_scope.html.
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[5] AM Palmieri, G M-Rigat, AK Srivastava, M Lewenstein, G R-Mielidzioć & M Płodzień, “Enhancing quantum state tomography via resource-efficient attention-based neural networks,” *Phys. Rev. Research* **6**, 033248 (2024).

AI Disclosure

OpenAI Codex was utilized for software architecture, verification tests, and schematic figures. Everything was checked and cross-verified by unit tests, exact-state identities, negative-log-likelihood monotonicity, cross-method comparisons, and document reviews by the team members. The scientific review, the compilation and execution of codes, data analysis, and the figure-generation were done by the team members.